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In a product test of a common cold remedy, x percent of the patients tested experienced side effects from the use of the drug and y percent experienced relief of cold symptoms. What percent of the patients tested experienced both side effects and relief of cold symptoms?

1. Of the 1,000 patients tested, 15 percent experienced neither side effects nor relief of cold symptoms.
2. Of the patients tested, 30 percent experienced relief of cold symptoms without side effects.

Algebra Sets

The Venn diagram on the following page represents the numbers of patients who experienced neither, one, or both side effects and relief. In the diagram,

$a = \text{number}$ who experienced side effects only,

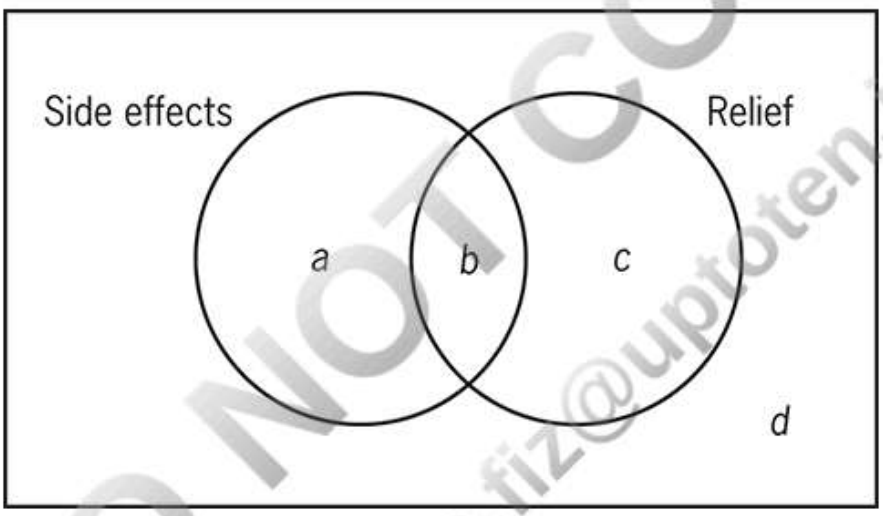
$b = \text{number}$ who experienced both,

$c = \text{number}$ who experienced relief only,

$d = \text{number}$ who experienced neither,

where $T = a + b + c + d$ is the total number of patients. Note that $a + b = \frac{x}{100}T$ and $b + c = \frac{y}{100}T$.

Determine the value of $\frac{b}{T}$.



1. Given that 1,000 patients were tested and (0.15)(1,000) = 150 patients experienced neither side effects nor relief, it follows that $a + b + c + d = 1{,}000$ and $d = 150$. Therefore, $a + b + c + 150 = 1{,}000$, or $a + b + c = 850$. However, there is no information about how the remaining 850 patients are assigned to the other categories. Thus, the percent of patients who experienced both side effects and relief could be any value between 0% and 85%; NOT sufficient.
2. Given that 30% of the patients experienced relief without side effects, there is no information about how the remaining 70% of the patients are assigned to the other categories. Thus, the percent of patients who experienced both side effects and relief could be any value between 0% and 70%; NOT sufficient.

Taking (1) and (2) together, it follows that 150 patients experienced neither side effects nor relief and (0.3)(1,000) = 300 patients experienced relief only. Therefore, from $a + b + c + d = T$ it follows that $a + b + 300 + 150 = 1{,}000$, or $a + b = 550$. However, it is still not possible to determine a unique value of $\frac{b}{T} = \frac{b}{1{,}000}$, the percent of patients who experienced both side effects and relief. For example, it is possible that $a = 300$ and $b = 250$ (that is, the percent who experienced both could be 25%), and it is possible that $a = 250$ and $b = 300$ (that is, the percent who experienced both could be 30%).

The correct answer is E; both statements together are still not sufficient.

•

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Three roommates—Bela, Gyorgy, and Janos—together saved money for a trip. The amount that Bela saved was equal to 8% of his monthly income. The amount that Gyorgy saved was exactly $\frac{1}{3}$ of the total amount saved by all 3 roommates. What was the total amount saved for the trip by all 3 roommates?

1. Bela had a monthly income of $2,000.
2. Janos saved 1.5 times as much for the trip as Bela.

Algebra First-Degree Equations

Letting B, G, and J represent the amounts saved by Bela, Gyorgy, and Janos, respectively, then $B = 0.08I$, where I represents Bela's monthly income, $G = \frac{1}{3}(B + G + J)$, and $G = \frac{1}{2}(B + J)$. Determine the value of $B + G + J$.

1. This allows the value of B to be determined but gives no information about the values of G or J; NOT sufficient.
2. This gives the relationship between the values of J and B but no information to determine the value of either B or J; NOT sufficient.

Taking (1) and (2) together, the values of both B and J can be determined, as can the value of G since $G = \frac{1}{2}(B + J)$. Thus, the value of $B + G + J$ can be determined.

The correct answer is C; both statements together are sufficient.

•

In each game of a certain tournament, a contestant either loses 3 points or gains 2 points. If Pat had 100 points at the beginning of the tournament, how many games did Pat play in the tournament?

1. At the end of the tournament, Pat had 104 points.
2. Pat played fewer than 10 games.

Arithmetic Computation with Integers

Pat either lost 3 points or gained 2 points in each game she played. Therefore, since she had 100 points at the beginning of the tournament, her score at the end of the tournament was $100 - 3x + 2y$, where x and y represent, respectively, the number of games in which she lost 3 points and the number of games in which she gained 2 points. Determine the value of $x + y$.

1. This indicates that $100 - 3x + 2y = 104$, but no information is given about the values of x or y. For example, the values of x and y could be 0 and 2, respectively, so that $x + y = 2$, or the values of x and y could be 2 and 5, respectively, so that $x + y = 7$; NOT sufficient.
2. This indicates that Pat played fewer than 10 games, but in each of the examples above, Pat played fewer than 10 games; NOT sufficient.

Since the examples given satisfy both (1) and (2), taking (1) and (2) together does not give enough information to determine the value of $x + y$.

The correct answer is E; both statements together are still not sufficient.

•

At the beginning of the year, the Finance Committee and the Planning Committee of a certain company each had n members, and no one was a member of both committees. At the end of the year, 5 members left the Finance Committee and 3 members left the Planning Committee. How many members did the Finance Committee have at the beginning of the year?

1. The ratio of the total number of members who left at the end of the year to the total number of members at the beginning of the year was 1:6.
2. At the end of the year, 21 members remained on the Planning Committee.

Algebra Applied Problems

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1. It is given that $\frac{8}{2n} = \frac{1}{6}$, since a total of 3 + 5 = 8 members left at the end of the year and there were a total of $n + n = 2n$ members at the beginning of the year. It follows that $(8)(6) = (2n)(1)$, or $n = 24$; SUFFICIENT.
2. It is given that $n - 3 = 21$, since 3 members from the original n members left the Planning Committee at the end of the year. It follows that $n = 21 + 3 = 24$; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

•

Some computers at a certain company are Brand X and the rest are Brand Y. If the ratio of the number of Brand Y computers to the number of Brand X computers at the company is 5 to 6, how many of the computers are Brand Y?

1. There are 80 more Brand X computers than Brand Y computers at the company.
2. There is a total of 880 computers at the company.

Algebra Simultaneous Equations

Let x be the number of Brand X computers and y be the number of Brand Y computers. We are given that $\frac{y}{x} = \frac{5}{6}$, or $6y = 5x$, and are to determine the value of y.

1. Given that $x = y + 80$, we have x solved in terms of y. Now replace x with $y + 80$ in the equation $6y = 5x$ to get an equation in which only y appears: $6y = 5(y + 80)$. This linear equation involving only y can be solved for a unique value of y, namely $y = 400$. Alternatively, $x = \left(\frac{6}{5}\right)y = y + \left(\frac{1}{5}y\right)$ and $x = y + 80$, so we have $\left(\frac{1}{5}y\right) = 80$, and hence $y = 400$; SUFFICIENT.
2. Given that $x + y = 880$, solve for x in terms of y to get $x = 880 - y$. Now replace x with $880 - y$ in the equation $6y = 5x$ to get an equation in which only y appears: $6y = 5(880 - y)$. This linear equation involving only y can be solved for a unique value of y, namely $y = 400$; SUFFICIENT.

Alternatively, in each of (1) and (2), we have two linear equations in two unknowns which, when graphed in the standard xy-coordinate plane, represent two lines having different slopes (hence the lines are not parallel), and therefore the two lines have exactly one point in common. The y-coordinate of that common point is the unique value of y that satisfies both equations.

The correct answer is D; each statement alone is sufficient.

•

Was the number of books sold at Bookstore X last week greater than the number of books sold at Bookstore Y last week?

1. Last week, more than 1,000 books were sold at Bookstore X on Saturday and fewer than 1,000 books were sold at Bookstore Y on Saturday.
2. Last week, less than 20 percent of the books sold at Bookstore X were sold on Saturday and more than 20 percent of the books sold at Bookstore Y were sold on Saturday.

Arithmetic Inequalities

Determine if Bookstore X sold more books last week than Bookstore Y.

1. This indicates that Bookstore X sold more books on Saturday than Bookstore Y, but it gives no information about the numbers of books sold by the two bookstores on the other days of last week; NOT sufficient.
2. This gives information about the percents of the books sold last week that were sold on Saturday, but it gives no information about the actual numbers of books sold at the two bookstores last week. Therefore, a comparison of the numbers of books sold last week by Bookstore X and Bookstore Y is not possible; NOT sufficient.

Taking (1) and (2) together, if x represents the number of books that Bookstore X sold on Saturday and T_X represents the total number of books that Bookstore X sold last week, then $x > 1{,}000$ and $x < 0.2T_X$. It follows that $1{,}000 < 0.2T_X$, and so $T_X > 5{,}000$. Similarly, if y represents the number of books that Bookstore Y sold on

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that $1{,}000 < 0.2T_X$, and so $T_X > 5{,}000$. Similarly, if y represents the number of books that Bookstore Y sold on Saturday and T_Y represents the total number of books that Bookstore Y sold last week, then $y < 1{,}000$ and $y > 0.2T_Y$. It follows that $0.2T_Y < 1{,}000$, and so $T_Y < 5{,}000$. Combining the inequalities gives $T_Y < 5{,}000 < T_X$, and so $T_Y < T_X$, which means that Bookstore X sold more books last week than Bookstore Y.

The correct answer is C; both statements together are sufficient.

•

From May 1 to May 30 in the same year, the balance in a checking account increased. What was the balance in the checking account on May 30?

1. If, during this period of time, the increase in the balance in the checking account had been 12 percent, then the balance in the account on May 30 would have been $504.
2. During this period of time, the increase in the balance in the checking account was 8 percent.

Arithmetic Applied Problems; Percents

1. Given that the amount on May 30 would have been $504 if the increase had been 12%, it follows that the amount on May 1 was $\$\frac{504}{1.12} = \$450$. However, the balance on May 30 cannot be determined because the (actual) percent increase is not given. For example, if the increase had been 10%, then the balance on May 30 would have been (1.1)($450) = $495, which is different from $504; NOT sufficient.
2. Given that the increase was 8%, the amount on May 30 could be $108 (if the amount on May 1 was $100), and the amount on May 30 could be $216 (if the amount on May 1 was $200); NOT sufficient.

Taking (1) and (2) together, it follows that the amount on May 30 was ($450)(1.08) = $486.

The correct answer is C; both statements together are sufficient.

•

A merchant discounted the sale price of a coat and the sale price of a sweater. Which of the two articles of clothing was discounted by the greater dollar amount?

1. The percent discount on the coat was 2 percentage points greater than the percent discount on the sweater.
2. Before the discounts, the sale price of the coat was $10 less than the sale price of the sweater.

Arithmetic Applied Problems; Percents

1. Given that the discount on the coat was 2 percentage points greater than the discount on the sweater, the sweater could have been discounted by the greater dollar amount ($30 sweater with 10% discount is a discount of $3; $20 coat with 12% discount is a discount of $2.40), and the coat could have been discounted by the greater dollar amount ($110 sweater with 10% discount is a discount of $11; $100 coat with 12% discount is a discount of $12); NOT sufficient.
2. Given that the coat's sale price was $10 less than the sweater's sale price, the same examples used to show that (1) is not sufficient can be used for (2); NOT sufficient.

Taking (1) and (2) together is of no more help than either (1) or (2) taken separately because the same examples used to show that (1) is not sufficient also show that (2) is not sufficient.

The correct answer is E; both statements together are still not sufficient.

•

If the positive integer n is added to each of the integers 69, 94, and 121, what is the value of n?

1. $69 + n$ and $94 + n$ are the squares of two consecutive integers.
2. $94 + n$ and $121 + n$ are the squares of two consecutive integers.

Algebra Computation with Integers

1. Given that $69 + n$ and $94 + n$ are squares of consecutive integers and differ by 25 (that is, $(94 + n) - (69 + n) = 25$), look for consecutive integers whose squares differ by 25. It would be wise to start with $10^2 = 100$ since $94 + n$ will be greater than 94.

$$10^2 = 100$$

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} difference is 21

$11^2 = 121$

} difference is 23

$12^2 = 144$

} difference is 25

$13^2 = 169$

Then the consecutive integers whose squares differ by 25 are $69 + n = 144$ and $94 + n = 169$. The value of n can be determined from either equation.

To be sure that the value for n is unique, consider algebraically that the difference of the squares of consecutive integers is 25. So, $(x+1)^2 - x^2 = 25$, from which it follows that $x^2 + 2x + 1 - x^2 = 25$ and $2x + 1 = 25$. This equation has a unique solution; SUFFICIENT.

2. Given that $94 + n$ and $121 + n$ are squares of consecutive integers and differ by 27 (that is, $(121 + n) - (94 + n) = 27$), look for consecutive integers whose squares differ by 27. It would be wise to start with $12^2 = 144$ since $121 + n$ will be greater than 121.

$12^2 = 144$

} difference is 25

$13^2 = 169$

} difference is 27

$14^2 = 196$

Then the consecutive integers whose squares differ by 27 are $94 + n = 169$ and $121 + n = 196$. The value of n can be determined from either equation; SUFFICIENT.

Tip: Since this and other such problems can be solved quickly using numbers, it might be helpful to have rapid recall of the perfect squares, possibly through 20^2.

The correct answer is D; each statement alone is sufficient.

•

If a merchant purchased a sofa from a manufacturer for $400 and then sold it, what was the selling price of the sofa?

1. The selling price of the sofa was greater than 140 percent of the purchase price.
2. The merchant's gross profit from the purchase and sale of the sofa was $\frac{1}{3}$ of the selling price.

Algebra Applied Problems

Letting s represent the selling price of the sofa, in dollars, determine the value of s.

1. The condition that the selling price of the sofa was greater than 140 percent of the purchase price is equivalent to $s > 1.4(400)$, from which it is not possible to determine the selling price; NOT sufficient.
2. Given that the gross profit from the purchase and sale of the sofa was $\frac{1}{3}$ of the selling price, it follows that $s - 400 = \frac{1}{3}s$, since gross profit is defined to be the selling price minus the purchase price. From this, a unique value of s can be determined; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

Last year, in a certain housing development, the average (arithmetic mean) price of 20 new houses was $160,000. Did more than 9 of the 20 houses have prices that were less than the average price last year?

1. Last year, the greatest price of one of the 20 houses was $219,000

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1. Last year, the greatest price of one of the 20 houses was $219,000.
2. Last year, the median of the prices of the 20 houses was $150,000.

Arithmetic Statistics

1. Given that the greatest of the prices was $219,000, the table below shows two possibilities in which the average price was $160,000.

Example I		Example II	
price	frequency	price	frequency
$101,000	1	$119,000	1
$160,000	18	$159,000	18
$219,000	1	$219,000	1

   In Example I, note that 101,000 + 219,000 = 320,000 = 160,000 + 160,000, so the sum of the prices is 20($160,000) and the average price is $160,000. Example II is then obtained from Example I by transferring $1,000 from each of the $160,000 prices to the $101,000 price, and thus the sum of the prices in Example II is also 20($160,000).

   Example I shows that only 1 of the prices could have been less than the average price, and Example II shows that 19 of the prices could have been less than the average price; NOT sufficient.

2. Given that the median of the 20 prices was $150,000, then when the prices are listed in numerical order from least to greatest, the average of the 10th and 11th prices is $150,000. Therefore, the 10th price must be less than or equal to $150,000, since if the 10th price was greater than $150,000, then both prices would be greater than $150,000 and their average would be greater than $150,000 (the median), contrary to what is given. Moreover, since the first 9 prices are less than or equal to the 10th price, it follows that the first 10 prices are each less than or equal to $150,000. Therefore, at least 10 of the prices are less than the average price; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

For a certain city's library, the average cost of purchasing each new book is $28. The library receives $15,000 from the city each year; the library also receives a bonus of $2,000 if the total number of items checked out over the course of the year exceeds 5,000. Did the library receive the bonus last year?

1. The library purchased an average of 50 new books each month last year and received enough money from the city to cover this cost.
2. The lowest number of items checked out in one month was 459.

Arithmetic Applied Problems

1. Given that the library purchased an average of 50 new books each month, for the entire year the library purchased a total of (50)(12) = 600 books for a total cost of (600)($28) = $16,800. Excluding any possible bonus, the library received $15,000 from the city. Since this amount received from the city is not enough to cover the cost of the books, and the information provided in (1) says that the total amount received from the city was enough to cover the cost of the books, it follows that the library received a bonus; SUFFICIENT.
2. Given that the least number of books checked out in one month was 459, it follows that the total number of books checked out for the year was at least (12)(459) = 5,508. Since this is greater than 5,000, it follows that the total number of books checked out for the year was greater than 5,000 and the library received a bonus; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

$$7, 9, 6, 4, 5, x$$

•

If x is a number in the list above, what is the median of the list?

1. $x > 7$
2. The median of the list equals the arithmetic mean of the list.

Arithmetic Statistics

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1. The given numbers in the list, in order, are 4, 5, 6, 7, 9. From (1), $x > 7$, so the list could be 4, 5, 6, 7, x, 9 or the list could be 4, 5, 6, 7, 9, x. In either case, the two middle numbers are 6 and 7 and the median is $\frac{6+7}{2} = \frac{13}{2}$; SUFFICIENT.
2. The mean of the list is $\frac{7+9+6+4+5+x}{6} = \frac{31+x}{6}$, which depends on the value of x. Likewise, the median depends on the value of x. If $x = 2$, then the list is 2, 4, 5, 6, 7, 9, which has median $\frac{5+6}{2} = \frac{11}{2}$ and arithmetic mean $\frac{31+2}{6} = \frac{33}{6} = \frac{11}{2}$, thereby satisfying (2). However, if $x = 8$, then the list is 4, 5, 6, 7, 8, 9, which has median $\frac{6+7}{2} = \frac{13}{2}$ and arithmetic mean $\frac{31+8}{6} = \frac{39}{6} = \frac{13}{2}$, thereby satisfying (2); NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

•

Three dice, each of which has its 6 sides numbered 1 through 6, are tossed. The sum of the 3 numbers that are facing up is 12. Is at least 1 of these numbers 5?

1. None of the 3 numbers that are facing up is divisible by 3.
2. Of the numbers that are facing up, 2, but not all 3, are equal.

Arithmetic Properties of Integers

When three dice, each with its 6 faces numbered 1 through 6, are tossed and the sum of the three integers facing up is 12, the possible outcomes are {1, 5, 6}, {2, 4, 6}, {2, 5, 5}, {3, 3, 6}, {3, 4, 5}, and {4, 4, 4}. Determine if, in the outcome described, at least one of the numbers is 5.

1. This indicates that none of the numbers is divisible by three. The outcome could be {4, 4, 4}. In this case, none of the three numbers is 5. On the other hand, the outcome could be {2, 5, 5}. In this case, at least one of the numbers is 5; NOT sufficient.
2. This indicates that two of the numbers, but not all three, are equal. The outcome could be {3, 3, 6}. In this case, none of the three numbers is 5. On the other hand, the outcome could be {2, 5, 5}. In this case, at least one of the numbers is 5; NOT sufficient.

Taking (1) and (2) together, of the possible outcomes, {1, 5, 6}, {2, 4, 6}, {3, 3, 6}, and {3, 4, 5} are eliminated because they do not satisfy (1), and {4, 4, 4} is eliminated because it does not satisfy (2). This leaves only {2, 5, 5}, and at least one number is 5.

The correct answer is C; both statements together are sufficient.

•

Mohit went cycling on 10 consecutive days. Every day he started from Point X, cycled a certain distance in an easterly direction, and then returned to Point X. What is the total distance he covered in those 10 days?

1. From the second day onward, Mohit covered twice the distance he covered the previous day. On the fifth day, he covered 3.2 km from Point X east, and then back to Point X.
2. On the first day, Mohit cycled less than 0.3 km.

Applied Problems

A man, Mohit, went cycling for 10 consecutive days. Each day he cycled east, starting from Point X, and back. We are asked to determine his total cycling distance over the 10 days.

1. On each day successively, Mohit cycled double the previous day's distance. On Day 5, he cycled 3.2 km from Point X east, and then back to Point X, and the total cycling distance on Day 4 was 1.6 km. For the final five days, the distance keeps doubling daily, so, for example, his cycling distance on Day 6 was 6.4 km. The information we have allows us to calculate the cycling distance on each of the 10 days. Finding the total distance covered in the 10 days is a simple matter of adding the distances for each of the 10 days. (1) is sufficient alone; SUFFICIENT.

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2. This information is not sufficient alone; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

•

Candidates A, B, and C competed in an election. This election usually comprises two rounds. In the first round, all three candidates compete. A candidate qualifies for the second round only if he/she receives at least 20% of the votes cast in the first round. If two candidates receive less than 20% of the votes cast, the other candidate wins the election by default. If a candidate receives more than 60% of the votes in the first round, he/she wins the election without the second round. Was there a second round of election?

1. The votes received by Candidates A and B together is 82% of the total votes cast.
2. The votes received by Candidates C and B together is 42% of the total votes cast.

Inference

An election had exactly three candidates A, B, and C competing. The election sometimes has two rounds of voting. A candidate who receives less than 20% of the votes cast is eliminated. If all but one candidate is eliminated, there is no second round. A candidate who receives 60% or more of the votes cast in the first round is automatically elected. The question posed is whether there was a second round in the election.

1. Candidates A and B together got 82% of the votes cast. That means that Candidate C was eliminated from any second round. However, we cannot decide whether another of the candidates was also eliminated; if so, the remaining candidate would have gotten more than 60% of the votes cast and been elected. But if Candidates A and B both got votes exceeding 20% of votes cast, there would be a runoff election. We do not have sufficient information to decide how the 82% of the votes split between Candidates A and B. So (1) does not provide information sufficient alone to decide whether there was a second round in the election; NOT sufficient.
2. Candidates C and B together got 42% of the total votes cast. That means that Candidate A got 58% of the votes cast. This means that nobody got 60% of the votes cast, and at least one of Candidate C or Candidate B got 20% or more of the votes cast—in either case, there would be a second round. So (2) is sufficient alone to determine that there was a second round in the election; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

There are a certain number of male and female students in a classroom. The weights of the male students range from 80 kg to 85 kg. The weights of the female students range from 70 kg to 75 kg. What is the average (arithmetic mean) weight of all the students in the classroom? All weights are rounded to the nearest 1 kg.

1. No two students weigh the same.
2. The number of male students is equal to the number of female students.

Applied Problems

In a classroom of male and female students, weights are recorded to the nearest kilo. Note that we are not told how many students are in the classroom.

1. No two students have the same recorded weight. This means that the number of males must be 6 or less, and similarly, the number of females must be 6 or less. However, we do not know the total number of students, nor the weights of the students—unless there are exactly 6 males and 6 females. Lacking more specific information, the mean weight of all students in the classroom cannot be calculated. So (1) is not sufficient alone; NOT sufficient.
2. The number of male students is equal to the number of female students. This does not help us calculate the total number of students in the classroom. And without this information, and without knowing the distribution of weights, we cannot calculate the mean of the students' weights. So (2) is not sufficient alone; NOT sufficient.

Neither (1) nor (2), whether alone or together, provides enough information about the number of students and distribution of weights.

The correct answer is E; both statements together are still not sufficient.

•

An alloy, Alloy K, is made by mixing certain quantities of iron and lead. The total weight of the alloy is 50 kg. What is the ratio of the weight of iron to the weight of lead in Alloy K?

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is the ratio of the weight of iron to the weight of lead in Alloy K?

1. If 4 kg of iron were to be replaced with 4 kg of lead, the percentage of lead, by weight, in Alloy K would become 40%.
2. The weight of iron per 100 cc is 0.7 times the weight of lead per 100 cc.

Applied Problems

We are asked to determine the ratio of iron and lead in an alloy weighing 50 kg. We are given a piece of hypothetical information concerning the alloy: what the concentration (i.e., percentage by weight) of lead in the alloy would be if 4 kg of iron were swapped for 4 kg of lead.

1. If 4 kg of iron in the alloy were to be swapped out for 4 kg of lead, the percentage of lead in the alloy would become 40%. In other words, 0.4 of the alloy would be lead. So there would be 20 kg of lead in the alloy. This means that before the hypothetical swap, there would have been 16 kg of lead and 34 kg of iron. So the ratio of iron to lead in the original alloy was 34:16. This means that (1) is sufficient alone to provide a definitive answer to the question asked; SUFFICIENT.
2. This information is simply a general fact about iron and lead and does not, by itself, have a bearing on the ratio of iron to lead in Alloy K. So this information is not sufficient alone to answer the question asked; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

•

A box contains exactly 40 bottles, each of which contains only one of three types of soda—orange, lemon, and ginger. What is the probability of randomly selecting a ginger soda, without any replacement in the box, on each of 3 consecutive attempts?

1. The number of orange sodas is 5 more than the number of ginger sodas.
2. The number of ginger sodas is 10 more than the number of lemon sodas.

Evaluation

A box of 40 sodas has 3 flavors, orange, ginger, and lemon, with a different number of sodas in each flavor. How probable is it to randomly select a ginger soda from the box on each of just three consecutive tries? To answer this question, here are pieces of information that will be needed: First, how many ginger sodas are in the box? Second, when a ginger soda is selected from the box, will it be removed and not replaced by a different flavor?

1. This information is not sufficient alone to determine the probability, since we don't know how many orange sodas are in the box; NOT sufficient.
2. This information is not sufficient alone to determine the number of ginger sodas in the box, since we don't know how many lemon sodas are in the box; NOT sufficient.

Using the information from (1) and (2) combined with the information in the stimulus, we can establish that the number of ginger sodas in the box is 15. When a ginger soda is selected on the first try, the number of ginger sodas left in the box is 14, and there is no addition to the box of a lemon or orange soda. With this information we can calculate the probability of choosing a ginger soda on the first two tries. Given similar information concerning non-replacement—i.e., that 2 ginger sodas chosen on the first two tries would not be replaced by two new sodas, ginger or other—then the probability of choosing 3 ginger sodas on consecutive tries can be calculated. (In that case, the result is 15/40 × 14/39 × 13/38.) So (1) and (2) are sufficient together.

The correct answer is C; both statements together are sufficient.

•

Workers A, B, and C, working at their individual constant rates, can work together and complete a piece of work in 6 days. How long will it take for Workers B and C alone to complete the work?

1. Worker A is twice as efficient as Worker B. Worker B is three times as efficient as Worker C.
2. The time taken by Worker A to do 9 units of work is the same as the time taken by Workers B and C together to do 6 units of work.

Applied Problems

1. This provides the ratio of the three amounts of work completed by each in that 6-day period: A:B:C = 6:3:1. This tells us that in 6 days, Worker A completes 1.5 times the work done by Workers B and C in that time. In other

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tells us that in 6 days, Worker A completes 1.5 times the work done by Workers B and C in that time. In other words, Workers B and C working together do 0.4 of all the work done, while Worker A does 0.6 of the work done. For Workers A and B together to do all of the work done by Workers A, B, and C, it will take 2.5 times 6 days, i.e., 15 days. So (1) is sufficient alone to infer an answer to the question asked; SUFFICIENT.

2. This provides that Worker A can do 9 units of work in the time that Workers A and B together do 6 units. That is, Worker A does 1.5 times the work that Workers B and C do together in the same time. In other words, we get the ratio in production A:(B+C) = 3:2. So the time it would take Workers B and C together to do all the work is (5/2 × 6) days, i.e., 15 days. So (2) is sufficient alone to infer an answer to the question asked; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

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The students in College X must take two, and no more, of these five subjects: computing, electronics, mechanics, architecture, and psychology. Students who take computing cannot take psychology. Students who take electronics cannot take architecture. Students who take mechanics cannot take electronics. Is Student G, a student at College X, taking psychology?

1. Student G is taking architecture as one of her two subjects.
2. Student G is taking neither electronics nor computing.

Inference

Every student at College X must take two and no more of the following five subjects: computing, electronics, mechanics, architecture, and psychology. The following constraints apply:

- Students taking computing cannot take psychology.
- Students taking electronics cannot take architecture.
- Students taking mechanics cannot take electronics.

Is Student G at College X taking psychology?

1. Student G is taking architecture as one of her two required subjects. This implies that she did not take electronics. Given the constraints, she would have taken psychology only if she was not taking computing. But we have no information that she is not taking computing. Therefore, (1) is not sufficient by itself to elicit a definitive answer to the question posed; NOT sufficient.
2. This provides that Student G is not taking computing. If she were, she would not be free to take psychology. Therefore, (2) is not sufficient alone to elicit a definitive answer to the question posed; NOT sufficient.

The information in (1) and (2) together with the stimulus is not sufficient to provide a definitive answer to the question. (1) tells us neither that Student G is taking psychology nor that she is not. (2) tells us that Student G is free to take psychology, but it tells us neither that she is nor is not taking it.

The correct answer is E; both statements together are still not sufficient.

•

A candy shop owner must mix three types of sweeteners—A, B, and C—in a certain ratio to get the desired mixture in his candies. What is the ratio of the weights of Sweetener A to Sweetener B to Sweetener C in the final mixture?

1. Sweeteners A, B, and C cost $40, $50, and $60 per pound. The average (arithmetic mean) cost of the final mixture is $50 per pound.
2. In the final mixture, the average (arithmetic mean) price of Sweeteners A and B together is $45, and the average (arithmetic mean) price of Sweeteners B and C together is $55.

Applied Problems

Three sweeteners A, B, and C are mixed as ingredients for candy. We are asked to determine the ratio among the weights of each based on information about cost per pound for each of the three ingredients and information about average (arithmetic mean) cost per pound for mixtures of two or three of the ingredients.

1. This information will not allow us to calculate weights of individual ingredients in the mixture, or even a ratio between individual weights. Therefore, (1) is not sufficient by itself; NOT sufficient.
2. This information will not allow us to calculate weights of individual ingredients in the mixture or even a ratio between the weights of individual ingredients. Therefore, (2) is not sufficient by itself; NOT sufficient.

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between the weights of individual ingredients. Therefore, (2) is not sufficient by itself, NOT sufficient.

If we combine the information provided in (1) and (2), we will be able to calculate the ratio between the weights of each of the three ingredients in the mixture.

Let a be the weight in pounds of Sweetener A, b the weight in pounds of Sweetener B, and c the weight in pounds of Sweetener C. From (1) and (2), we get the following equations:

$(40a + 50b) = 45(a + b)$

$(50b + 60c) = 55(b + c)$.

From the first, we can infer $5b - 5a = 0$, and therefore that $a = b$. Similarly, from the second, we can infer that $b = c$. So $a : b : c = 1 : 1 : 1$. So (1) and (2) are sufficient together.

The correct answer is C; both statements together are sufficient.

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A movie theater increased the ticket price for evening shows by 10% and decreased the ticket price for morning shows by 40%. The

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